

YEAR 11
TERM 4 EXAMINATION 2010

MATHEMATICS
EXTENSION 1

Time Allowed – 90 minutes
(Plus 5 minutes Reading Time)

All questions may be attempted

All questions are of equal value

Department of Education approved calculators are permitted

In every question, show all necessary working

Marks may not be awarded for careless or badly arranged work

No grid paper is to be used unless provided with the examination paper

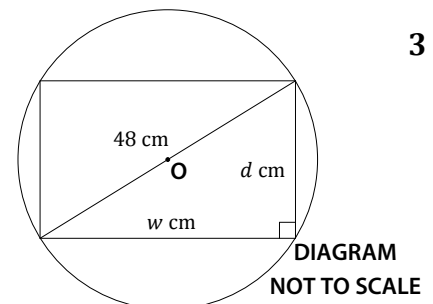
The answers to all questions are to be returned in separate bundles clearly labelled Question 1, Question 2, etc. Each question must show your Candidate's Number.

Question 1 (9 Marks)**Marks**

- (a) Given $A(-7, 1)$, $B(-4, 6)$, $C(1, 3)$ and $D(-2, -2)$:
- Prove that AC and BD are equal in length. **1**
 - Hence or otherwise prove that $ABCD$ is a square. **2**
- (b) Evaluate $\operatorname{cosec}\left(\frac{-5\pi}{6}\right)$. **1**
- (c) Find $\int (5 - 4x)^6 dx$. **1**
- (d) Find the equation for y , given that $y' = \frac{x+1}{x}$ and y passes through $(1, 5)$. **2**
- (e) Solve for x : $\frac{2x}{1-x} < 0$. **2**

Question 2 (9 Marks) – START A NEW PAGE

- (a) Find the equation of the inflexional tangent of $y = x^3 - 2x^2 - 2x + 2$. **3**
- (b) Using the substitution $u = x^3 - 2$, evaluate the integral $\int_2^3 x^2 \sqrt{x^3 - 2} dx$. **3**
- (c) The strength s of a rectangular beam is in proportion to the product of its width w and the square of its depth d , that is, $s = kwd^2$ for some positive constant k . Find the dimensions of the strongest rectangular beam that can be cut from a cylindrical log of diameter 48cm. **3**

**Question 3 (9 Marks) – START A NEW PAGE**

- (a) Determine the minimum value of $y = \frac{x+1}{x^2}$. **3**
- (b) Use Simpson's rule with 3 function values to estimate $\int_{-1}^3 \sqrt{x+3} \ln \sqrt{x^2+1} dx$ correct to 2 decimal places. **2**
- (c) Given $y = \frac{\tan x}{\sqrt{3}} - 1$:
- Prove that $(0, -1)$ is a point of inflexion. **2**
 - Hence, neatly sketch the curve for $0 \leq x \leq 2\pi$. **2**

Question 4 (9 Marks) – START A NEW PAGE**Marks**

- (a) Find $\int \frac{2x^2-6}{x^2+9} dx$. **2**
- (b) By considering the equation $x^2 + y^2 = r^2$, prove that the volume of a sphere with radius r is given by $V = \frac{4}{3}\pi r^3$. **3**
- (c) The region bounded by the curve $y = \frac{1}{x}$ and the x -axis between $x = 1$ and $x = 2$ is rotated through one complete revolution about the y -axis. Find the exact volume of the solid of revolution so formed. **4**

Question 5 (9 Marks) – START A NEW PAGE

- (a) If $f(x) = \sin^{-1} x$ & $g(x) = \cos^{-1} x$, prove that $\frac{d}{dx}[f(x)g(x)] = -f'(x)[f(x) - g(x)]$. **2**
- (b) A function is defined by $f(x) = \frac{\sqrt{4-x^2}}{x}$.
- Find the largest positive domain for which $y = f(x)$ has an inverse. **1**
 - For the domain in (i), find an expression for $y = f^{-1}(x)$, the inverse of $y = f(x)$. **2**
 - State the domain and range of $y = f^{-1}(x)$. **2**
 - For the domain in (i) and on the same graph, neatly sketch $y = f(x)$ and $y = f^{-1}(x)$, clearly identifying each sketch. **2**

EXAM CONTINUES OVERLEAF

Question 6 (9 Marks) – START A NEW PAGE**Marks**

- (a) i. Express $3 \sin x + \sqrt{3} \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and $0 \leq \alpha \leq 2\pi$. 2
- ii. Hence find the general solutions to $3 \sin x + \sqrt{3} \cos x = 3$. 2
- (b) $ABCD$ is a square with sides of length 1m. Quadrants CBD and APQ touch at point X . CP and CQ are constructed such that the shaded area is enclosed by figure $RQXPS$.

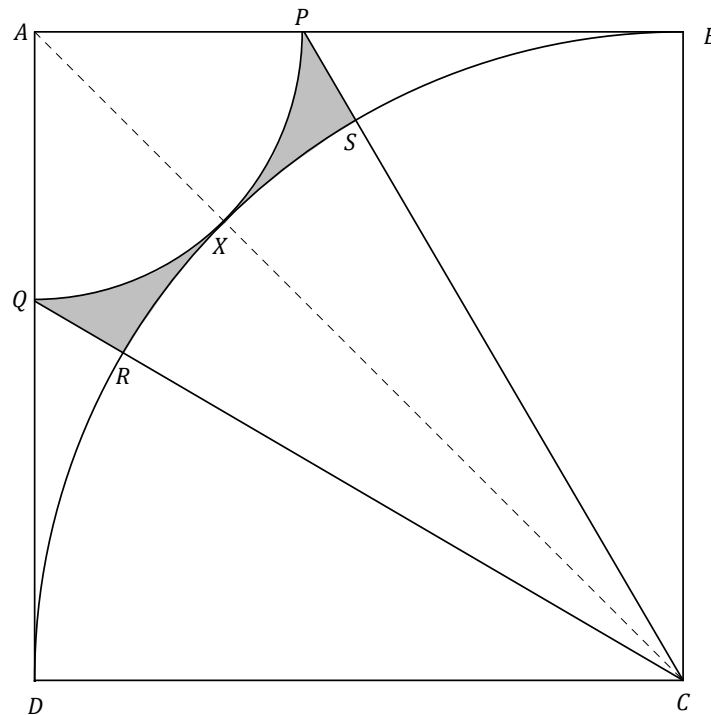


DIAGRAM
NOT TO SCALE

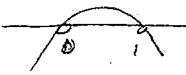
- i. Show that $\angle PCQ = 0.51$ radians (to 2 decimal places). 2
- ii. Hence or otherwise calculate the shaded area to the nearest square centimetre. 3

Question 7 (9 Marks) – START A NEW PAGE

- (a) Find the equation of the normal to the curve $y = \sin^{-1}\left(\frac{\sqrt{x}}{2}\right)$ where $x = 1$. 3
- (b) For the function $y = \tan^{-1}(2x) + \tan^{-1}\left(\frac{2}{x}\right)$:
- i. Show that $\frac{dy}{dx} = \frac{2}{4x^2+1} - \frac{2}{4+x^2}$. 1
- ii. State the domain and range of the function. 2
- iii. Neatly sketch the graph of $y = \tan^{-1}(2x) + \tan^{-1}\left(\frac{2}{x}\right)$, clearly labelling all stationary points. 3

END OF EXAM

Y12 M. Ext 1 Assess TEST 1
TERM 4, 2020

MATHEMATICS Extension 1 : Question.....1		
Suggested Solutions	Marks	Marker's Comments
a) $AC = \sqrt{(-7-1)^2 + (1-3)^2} = \sqrt{8^2 + 2^2} = \sqrt{68}$ $BD = \sqrt{(-4+2)^2 + (6-2)^2} = \sqrt{2^2 + 8^2} = \sqrt{68}$ $\therefore AC = BD$	1	
ii) $m(AC) = \frac{3-1}{1-7} = \frac{2}{-6} = -\frac{1}{3}$ $m(BD) = \frac{6-2}{-4-2} = \frac{4}{-6} = -\frac{2}{3}$ $\therefore m(AC) \times m(BD) = -1 \quad AC \perp BD$	1	
$mid(AC) = \left(\frac{-7+1}{2}, \frac{1+3}{2}\right) = (-3, 2)$ $mid(BD) = \left(\frac{-4+2}{2}, \frac{6-2}{2}\right) = (-1, 2)$ diagonals are equal and bisect each other at 90°	1	most students forgot to prove bisect only 1 m must give reason $-\frac{1}{2}m$
b) $\frac{1}{\sin\left(-\frac{5\pi}{6}\right)} = \frac{1}{-\frac{1}{2}} = -2$	1	no half mark. Overall, well done.
c) $\frac{(5-4x)^7}{-28} + c$	1	
d) $y = \int 1 + \frac{1}{x} dx = x + \ln x + c$ $5 = 1 + \ln 1 + c$ $c = 4$ $y = x + \ln x + 4$	1	forgot last line $-\frac{1}{2}m$
e) $\frac{2x}{1-x} > (1-x)^2 < 0(1-x)^2$ $2x(1-x) < 0$ $\therefore x < 0 \text{ or } x > 1$	1	
	1	$x < 0 \text{ or } x > 1$ use 'and' $-\frac{1}{2}m$

4R II TERM 4 MATHEMATICS Extension 1: Question 2 2010		
Suggested Solutions	Marks	Marker's Comments
(c) $d^2 + w^2 = 48^2$ (Pythagoras theorem) $d^2 = 48^2 - w^2$ $\therefore S = kw(48^2 - w^2)$ $S = 48^2 kw - kw^3$ $\therefore \frac{dS}{dw} = 48^2 k - 3kw^2$ $= k(48^2 - 3w^2)$ Stationary pts exist when $\frac{dS}{dw} = 0$ $\therefore k(48^2 - 3w^2) = 0$ $3w^2 = 48^2$ $w^2 = \frac{48^2}{3}$ $w = \frac{48}{\sqrt{3}} \quad (\text{as } w > 0)$ $w = 16\sqrt{3} \text{ cm } (\sqrt{168})$ Test nature $\frac{d^2S}{dw^2} = -6kw$ When $w = 16\sqrt{3}$, $\frac{d^2S}{dw^2} = -96\sqrt{3}k$ Now for $k > 0$ $\frac{d^2S}{dw^2} < 0$ concave down $\therefore$ local maximum exists when $w = 16\sqrt{3} \text{ cm}$ As the function is continuous for $0 \leq w \leq 48$ and there is only one turning pt. the local max. is also the absolute max. When $w = 16\sqrt{3}$, $d^2 = 48^2 - (16\sqrt{3})^2$ $d^2 = 1536$ $d = 16\sqrt{6} \text{ cm } (d > 0)$ $\therefore$ Dimensions of the strongest rectangular beam are $16\sqrt{3} \text{ cm}$ wide and $16\sqrt{6} \text{ cm}$ deep.	$\frac{1}{2}$ $\frac{1}{2}$ 1 1	Not squaring 48 made the calculations a lot easier, so a max. of 2 marks for the whole question. $\frac{1}{2}$ mark deducted for not stating $w > 0$ & $d > 0$ Locating the stat. pt. If the students did the 1st derivative test they needed to say $k > 0$ for the full mark. (Strength is positive) $\frac{1}{2}$ mark deducted for not proving absolute max. $\frac{1}{2}$ mark deducted for no units

4R11 TERM 4 MATHEMATICS Extension 1: Question 2 2010

Suggested Solutions	Marks	Marker's Comments								
<u>Q2</u> (a) $y = x^3 - 2x^2 - 2x + 2$ $y' = 3x^2 - 4x - 2$ $y'' = 6x - 4$ For inflexion pts $y'' = 0$ $6x - 4 = 0$ $x = \frac{2}{3}, y = \frac{2}{27}$ <table border="1"> <tr> <td>x</td><td>$\frac{1}{3}$</td><td>$\frac{2}{3}$</td><td>1</td></tr> <tr> <td>y''</td><td>-2</td><td>0</td><td>2</td></tr> </table> Change in concavity $\therefore$ Inflexion pt. at $(\frac{2}{3}, \frac{2}{27})$ When $x = \frac{2}{3}, y' = -\frac{10}{3}$ Eqn. of inflexional tangent is: $y - \frac{2}{27} = -\frac{10}{3}(x - \frac{2}{3})$ $y - \frac{2}{27} = -\frac{10x}{3} + \frac{20}{9}$ $27y - 2 = -90x + 60$ $90x + 27y - 62 = 0$	x	$\frac{1}{3}$	$\frac{2}{3}$	1	y''	-2	0	2	1 $\frac{1}{2}$ $\frac{1}{2}$ 1	$\frac{1}{2}$ mark deducted for not testing for inflexion pt. $y = -\frac{10}{3}x + 2\frac{8}{27}$ $y = -\frac{10x}{3} + \frac{62}{27}$
x	$\frac{1}{3}$	$\frac{2}{3}$	1							
y''	-2	0	2							
(b) $\int_2^3 x^2 \sqrt{x^3 - 2} dx = \int_6^{25} \frac{1}{3} \sqrt{u} du$ Let $u = x^3 - 2$ $\frac{du}{dx} = 3x^2$ $du = 3x^2 dx$ When $x = 2, u = 6$ $x = 3, u = 25$ $\therefore \int_6^{25} \frac{1}{3} \sqrt{u} du = \frac{1}{3} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_6^{25}$ $= \frac{2}{9} \left[u^{\frac{3}{2}} \right]_6^{25}$ $= \frac{2}{9} (12.5 - 6\sqrt{6})$ <u>OR</u> $= \frac{250 - 12\sqrt{6}}{9}$ <u>OR</u> $\approx 24.51 \text{ (2dp)}$	1 1 1	1 mark for correct substitution 1 mark for the change of limits 1 mark for correct answer								

MATHEMATICS Extension 1: Question 3

Suggested Solutions

Marks

Marker's Comments

Q3(a) $y = \frac{x+1}{x} = x^{-1} + x^{-2}, x \neq 0$

$\frac{dy}{dx} = -\frac{1}{x^2} - \frac{2}{x^3} = -\frac{(x+2)}{x^3}$

$\frac{d^2y}{dx^2} = \frac{2}{x^3} + \frac{6}{x^4} = \frac{2x+6}{x^4}$

For possible TPs to occur $\frac{dy}{dx} = 0$

$\therefore -\frac{1}{x^2} - \frac{2}{x^3} = -\frac{(x+2)}{x^3} = 0$

$\therefore x = -2, y = -\frac{1}{4}$ SP = $(-2, -\frac{1}{4})$

TEST: at $x = -2$

$\frac{d^2y}{dx^2} = \frac{2}{8} + \frac{6}{16} = \frac{1}{8} > 0$

$\therefore$ concave upwards

$\therefore$ a Rel. min TP at $x = -2$

Since y is cont. and differentiable for $x < 0$ or $x > 0$ and there is only one TP

$\therefore$ absolute min is $-\frac{1}{4}$ (when $x = -2$)

OR: $x^2y = x+1$

$x^2y - x - 1 = 0$

For x to be real $\Delta \geq 0$ DF

$\therefore (-1)^2 - 4 \times y \times (-1) \geq 0$

$1 + 4y \geq 0$

$y \geq -\frac{1}{4}$

$\therefore R_y = \{y: y \geq -\frac{1}{4}\}$

when $y = -\frac{1}{4}$ there is a TP at $x = \frac{-(-1)}{2 \times (-\frac{1}{4})} = -2$

$\therefore$ min (abs) value is $-\frac{1}{4}$ when $x = -2$

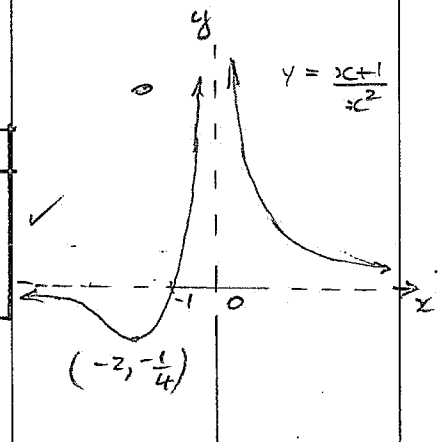
x	-3	-2	-1.9	-1
$\frac{dy}{dx}$	$-\frac{1}{27}$	0	0.0145	1
	-	0	+	

$\frac{1}{2}$

$\frac{1}{2}$ ea.

$\frac{1}{2}$

Q. Rule given
 $\frac{1x^2 - (x+1) \times 2x}{x^4} = \frac{-x^2 - 2x}{x^4}$
=



(b) $\int_{-1}^3 \sqrt{x+3} \ln \sqrt{x^2+1} dx$: $h = \frac{3 - (-1)}{2} = \frac{4}{2} = 2$

$\approx \frac{h}{3} [y_1 + 4y_2 + y_3]$

x	-1	1	3
y	$\sqrt{2} \ln \sqrt{2}$ 0.4901...	$2 \ln \sqrt{2}$ 0.6931...	$\sqrt{6} \ln \sqrt{10}$ 2.8201...
	y_1	y_2	y_3

$= \frac{2}{3} [\sqrt{2} \ln \sqrt{2} + 4 \times 2 \ln \sqrt{2} + \sqrt{6} \ln \sqrt{10}]$

$= \frac{2}{3} [6.082797...] = 4.055198052...$

$= 4.06$ (2 d.p.)

$\frac{1}{2}$

$\frac{b-a}{6} = \frac{3 - (-1)}{6} = \frac{2}{3}$

$a = -1$

$b = 3$

$\frac{a+b}{2} = 1$

$\frac{1}{2}$ For $\frac{2}{3}$

1 For CNE y_1, y_2, y_3

$\frac{1}{2}$ For 4.06

gained correctly



MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments

(c) (i) $y = \frac{1}{\sqrt{3}} \tan x - 1$

$$\frac{dy}{dx} = \frac{1}{\sqrt{3}} \sec^2 x$$

$$\frac{d^2y}{dx^2} = \frac{2}{\sqrt{3}} \sec^2 x \tan x = \frac{2}{\sqrt{3}} \frac{\sin x}{\cos^3 x}$$

For possible P.O.I.s to occur $\frac{d^2y}{dx^2} = 0$

$$\therefore \sec^2 x \tan x = 0 \text{ i.e. } \tan x = 0$$

$$\therefore x = 0, \pi, 2\pi, \dots$$

$$y = -1, -1, -1, \dots$$

$\therefore$ POSS POI $(0, -1)$

TEST

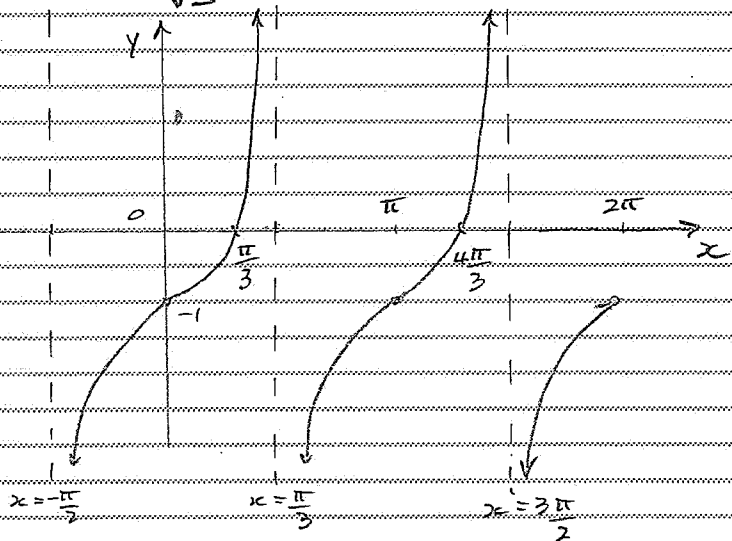
x	-1	0	0.1	$\frac{1}{2}$	1	$\frac{\pi}{4}$
$\frac{d^2y}{dx^2}$	-6.16	0	0.117	0.819	6.16	$\frac{4}{\sqrt{3}}$
	$\ominus$					

$\therefore$ a change in sign

$\therefore$ a change in concavity $(- \rightarrow +)$

$\therefore$ a Point of inflection at $(0, -1)$

(ii) Sketch $y = \frac{1}{\sqrt{3}} \tan x - 1$ $-\frac{\pi}{2} \leq x \leq 2\pi$



$(\frac{1}{2})$

$\frac{1}{2}$

or substit $x=0$
in $\frac{dy}{dx}$ to get $y'' = 0$

$$y'' = \frac{2}{\sqrt{3}} \sec^2 x \tan x$$

x	0^-	0	0^+
y''	$\frac{2(+)(-)}{\sqrt{3}}$	0	$\frac{2(+)(+)}{\sqrt{3}}$

1 For Test

2

$\frac{1}{2}$ For x -int $\frac{\pi}{3}$ or $\frac{4\pi}{3}$

$\frac{1}{2}$ POI at $(0, -1)$

$\frac{1}{2}$ V.A $x = \frac{\pi}{2}; \frac{3\pi}{2}$

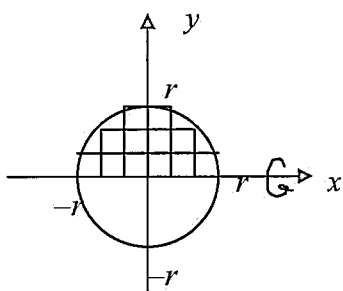
$\frac{1}{2}$ shape

2

$$\begin{aligned}
 \text{(a)} \quad \int \frac{2x^2 - 6}{x^2 + 9} dx &= 2 \int \frac{x^2 + 9 - 12}{x^2 + 9} dx \\
 &= 2 \int 1 dx - \int \frac{12}{x^2 + 9} dx \quad \checkmark \\
 &= 2 \left[x - 4 \tan^{-1} \left(\frac{x}{3} \right) \right] + c, \quad \checkmark \\
 &\text{where } c \text{ is a constant}
 \end{aligned}$$

- Correctly breaks up the fraction – 1 mk
- Correctly integrates both – 1 mk
- Correctly integrating $\int \frac{1}{x^2 + 9} dx$ - ½ mark only

$$\text{(b)} \quad x^2 + y^2 = r^2 \rightarrow \text{circle centre } (0,0) \text{ radius } r.$$

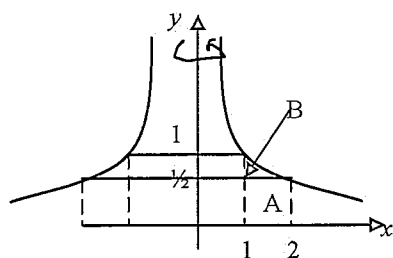


Take shaded area and rotate it around the x – axis

$$\begin{aligned}
 \therefore \text{volume} &= \pi \int_{-r}^r y^2 dx \\
 &= \pi \int_{-r}^r (r^2 - x^2) dx \quad \checkmark \\
 &= \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r \quad \checkmark \\
 &= \pi \left[r^2(r) - \frac{(r)^3}{3} - r^2(-r) + \frac{(-r)^3}{3} \right] \\
 &= \pi \left[2r^3 - \frac{2r^3}{3} \right] = \frac{4\pi r^3}{3} \text{ as required} \\
 &\quad \checkmark
 \end{aligned}$$

- Students can do rotation about the y – axis also.
- Correctly uses the equation of circle to get to correct definite integral – 1 mk
- Correctly integrates wrt x – 1 mk
- Correctly substitutes and simplifies clearly showing how they get answer – 1 mk

(c)



To find this volume you must add the Vol A to Vol B

$\therefore$ Volume A = Vol. cylinder

$$= \pi(2^2)\left(\frac{1}{2}\right) - \pi(1^2)\frac{1}{2} = \frac{3\pi}{2}$$

$$\text{Volume B} = \pi \int_{\frac{1}{2}}^1 \frac{1}{y^2} dy - \pi(1^2)\frac{1}{2}$$

$$= \pi \left[-\frac{1}{y} \right]_{\frac{1}{2}}^1 - \frac{\pi}{2}$$

$$= -\pi + 2\pi - \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

$$\therefore \text{ total volume} = \frac{3\pi}{2} + \frac{\pi}{2} = 2\pi \text{ cubic units}$$

TOTAL = 9 MARKS

- This question was very poorly done.
- Most students had no idea what to do with the question.

- Correctly finds vol. A
1 ½ mks

- Correctly integrates

$$\pi \int_{\frac{1}{2}}^1 \frac{1}{y^2} dy - 1 \frac{1}{2} \text{ mks}$$

- Correctly subtracts off the vol. of cylinder –
½ mk

- Final answer – ½ mk

NOTES:

1. Max 1 mk if correctly did rotation about x – axis i.e.

$$\pi \int_1^2 \frac{1}{y^2} dx$$

2. Also some students only did

$$\pi \int_1^2 \frac{1}{y^2} dy - \text{again max 1 mk.}$$

MATHEMATICS Extension 1 : Question... 5

Suggested Solutions

Marks

Marker's Comments

(a) $y = \sin^{-1} x \cdot \cos^{-1} x$

$f'(x) = \frac{1}{\sqrt{1-x^2}} \quad g'(x) = \frac{-1}{\sqrt{1-x^2}}$

$\frac{dy}{dx} = \sin^{-1} x \cdot \frac{-1}{\sqrt{1-x^2}} + \frac{\cos^{-1} x}{\sqrt{1-x^2}}$

$= \frac{-1}{\sqrt{1-x^2}} [\sin^{-1} x + \cos^{-1} x]$

$= -f'(x) [f(x) - g(x)]$

1/2

1/2

1/2

1/2

(b) (i) Domain: $0 < x \leq 2$

1/2 for "<"

1/2 for 0 and 2

(ii) $y = \frac{\sqrt{4-x^2}}{x}$

$x = \frac{\sqrt{4-y^2}}{y}$

$xy = \sqrt{4-y^2}$

$x^2 y^2 = 4 - y^2$

$x^2 y^2 + y^2 = 4$

$y^2 (x^2 + 1) = 4$

$y^2 = \frac{4}{x^2 + 1}$

$y = \frac{\pm 2}{x^2 + 1}$

$y = \frac{2}{x^2 + 1}$ but Range is $0 < y \leq 2$ for the inverse function!!

1/2

1/2

1/2

(iii) D: $x > 0$

R: $0 < y \leq 2$

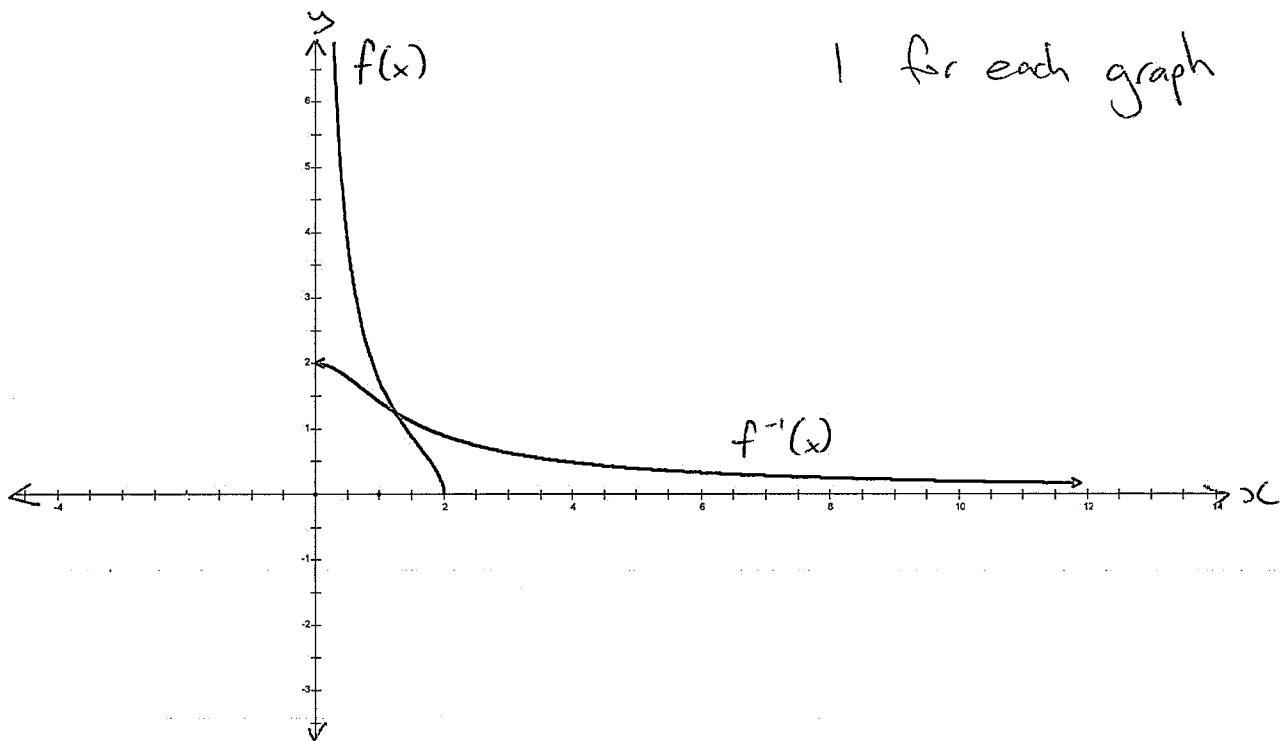
1

1

(iv)

lots of students wrote the domain and range wrong but then managed to graph the correct graph!???

1 for each graph



Suggested Solutions

Marks

Marker's Comments

$$R \sin(x-d) = R \sin x \cos d - R \cos x \sin d = 3 \sin x + \sqrt{3} \cos x$$

$$\therefore R \sin d = -\sqrt{3} \quad (1) \quad R \cos d = 3 \quad (2) \quad \sin d < 0 \quad \cos d > 0$$

$$\tan d = \frac{-1}{\sqrt{3}} \quad \frac{3\pi}{2} < d < 2\pi$$

$$d = \frac{11\pi}{6}$$

$$R = \sqrt{3^2 + (\sqrt{3})^2} =$$

$$= \sqrt{12}$$

$$= 2\sqrt{3}$$

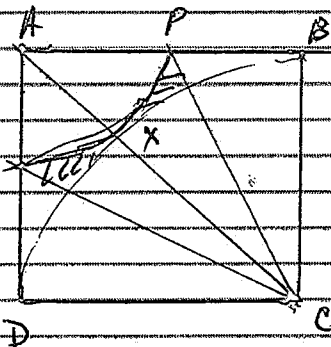
$$\therefore 3 \sin x + \sqrt{3} \cos x = 2\sqrt{3} \sin\left(x - \frac{11\pi}{6}\right)$$

$$(ii) \therefore 2\sqrt{3} \sin\left(x - \frac{11\pi}{6}\right) = 3$$

$$\sin\left(x - \frac{11\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$x - \frac{11\pi}{6} = \pi k + (-1)^k \cdot \frac{\pi}{3}$$

$$x = \frac{11\pi}{6} + \pi k + (-1)^k \cdot \frac{\pi}{3} \quad k \text{ integer}$$



$$AC = \sqrt{1^2 + 1^2} = \sqrt{2} \quad (\text{Pythagoras})$$

$$\therefore AQ = \sqrt{2} - 1$$

$$\therefore AP = \sqrt{2} - 1$$

$$DQ = \sqrt{2} - (\sqrt{2} - 1) = 2 - \sqrt{2}$$

$$\tan \angle QCD = \frac{2 - \sqrt{2}}{1}$$

$$\angle QCD = \tan^{-1}(2 - \sqrt{2})$$

$$\angle PCQ = \frac{\pi}{2} - 2 \tan^{-1}(2 - \sqrt{2})$$

$$= 0.51 \text{ rad}$$

$$\text{Area } \triangle PCQ = 2 \times \frac{1}{2} \times (2 - \sqrt{2}) \times 1$$

$$= 2 \times \frac{1}{2} \times (\sqrt{2} - 1) \times 1$$

$$= \sqrt{2} - 1 = 0.41421$$

$$\text{Area quadrant APQ} = \frac{1}{2} \times (\sqrt{2})^2 \times \frac{\pi}{2} = 0.13475$$

$$\text{Area sector QRS} = \frac{1}{2} \times 1^2 \times 0.51 = 0.25550$$

$$\text{Total} = \text{Area } \triangle PCQ - \text{Area quadrant APQ} - \text{Area sector QRS}$$

$$= 0.41421 - 0.13475 - 0.25550$$

$$= 0.02396$$

$$= 0.0240 \text{ (m}^2\text{)}$$

$$\text{OR } 240 \text{ cm}^2 \text{ (nearest cm}^2\text{)}$$

Q7

$$\begin{aligned}
 (a) \quad y &= \sin^{-1}\left(\frac{\sqrt{x}}{2}\right) \\
 y' &= \frac{1}{\sqrt{1 - \left(\frac{\sqrt{x}}{2}\right)^2}} \cdot \frac{1}{2} \times \frac{1}{2} x^{-\frac{1}{2}} \\
 &= \frac{1}{4\sqrt{x}\sqrt{1 - \frac{x}{4}}} \\
 &= \frac{1}{4\sqrt{x} \cdot \frac{1}{2}\sqrt{4-x}} \\
 &= \frac{1}{2\sqrt{x}\sqrt{4-x}} \\
 \therefore m_{\text{tangent}} &= y'(1) = \frac{1}{\sqrt{12}} \\
 \therefore m_{\text{normal}} &= -\sqrt{12}
 \end{aligned}$$

When $x = 1$, $y = \sin^{-1}\frac{1}{2} = \frac{\pi}{6}$

Hence equation of normal is

$$y - \frac{\pi}{6} = -\sqrt{12}(x - 1)$$

$$\Rightarrow y = -\sqrt{12}x + (\sqrt{12} + \frac{\pi}{6})$$

$\frac{1}{2}$ for derivative (in unsimplified form)

$\frac{1}{2}$ for slope of tangent

$\frac{1}{2}$ for slope of normal

$\frac{1}{2}$ for coordinates of point of contact

$\frac{1}{2}$ for m and coordinates in point gradient formula

$\frac{1}{2}$ for expressing equation either in the form

$$y = mx + b \text{ or } ax + by + c = 0$$

Note: (i) If m_{tangent} used to get equation, then max $2\frac{1}{2}$

(ii) If derivative incorrect, then max $2\frac{1}{2}$

(iii) If no working for $\pi/6$ minus $\frac{1}{2}$

(iv) Minus $\frac{1}{2}$ for poor notation eg $m_N, m_T, \perp y'$, perpendicular -1 etc

(b) (i) $y = \tan^{-1} \frac{x}{2} + \tan^{-1} \frac{2}{x}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1 + (2x)^2} \cdot 2 + \frac{1}{1 + \left(\frac{2}{x}\right)^2} \cdot (-2x^{-2}) \\ &= \frac{2}{1 + 4x^2} - \frac{2}{x^2(1 + \frac{4}{x^2})} \\ &= \frac{2}{1 + 4x^2} - \frac{2}{x^2 + 4} \end{aligned}$$

$\frac{1}{2}$ for both terms of derivative

$\frac{1}{2}$ for simplification - must show how.

(ii) Domain: $x \neq 0, x \in \mathbb{R}$ and

Range: $\frac{\pi}{2} < y \leq 2.21 \cup -2.21 \leq y < -\frac{\pi}{2}, y \in \mathbb{R}$

$x \neq 0$ or $x < 0$ and $x > 0$
then 1 mark

$\frac{1}{2}$ for $x \in \mathbb{R}$ only
 0 for $y \in \mathbb{R}$ only.
 $\frac{1}{2}$ for each range

(iii) $\frac{dy}{dx} = 0$... for stationary points:

i.e.

$$\frac{2}{1 + 4x^2} - \frac{2}{x^2 + 4} = 0$$

$\Rightarrow$

$$\frac{2}{1 + 4x^2} = \frac{2}{x^2 + 4}$$

$$x^2 + 4 = 1 + 4x^2$$

$$x = \pm 1$$

when $x = 1, y = 2.21$ or $2 \tan^{-1} 2$

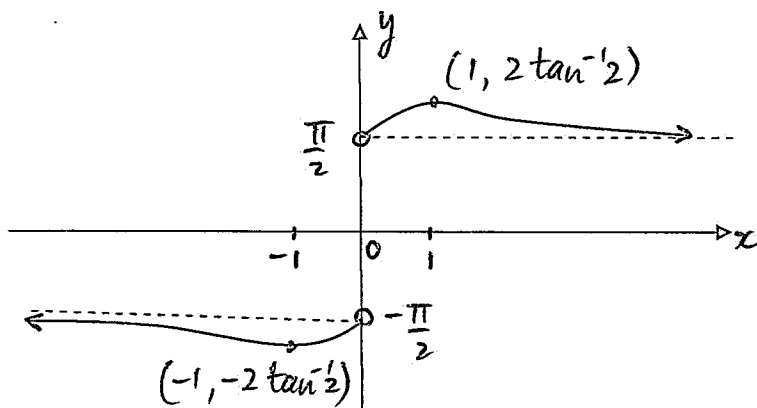
when $x = -1, y = -2.21$ or $2 \tan^{-1}(-2)$

$\frac{1}{2}$ for stationary points

x	-1.5	-1	-0.5	0.5	+1	1.5
y'	-3/25	0	9/17	9/17	0	-3/25
	\	—	/	/	—	\

} $\frac{1}{2}$ for testing

Hence $(1, 2 \tan^{-1} 2)$ is a local maximum and $(-1, -2 \tan^{-1} 2)$ is a local minimum



$\frac{1}{2}$ for correct intercepts

$\frac{1}{2}$ for shape, including discontinuity and asymptote

$\frac{1}{2}$ for max on graph

$\frac{1}{2}$ for min on graph

max $1\frac{1}{2}$ for

with correct TP + working